

AASRA: Precision Agronomic Platform

How the 9 Core Features Work — Architectural & Operational Guide

Document Target: Agronomists, Developers, Product Evaluators & Farmers | **Standard:** ICAR NBSS&LUP; & CIBRC Agronomic Protocols | **Version:** 2.4 (Production Verified)

1. Executive Summary & Core Philosophy

Modern agriculture often suffers from generic, one-size-fits-all recommendations. In reality, two neighbouring farmers growing the same crop in the same village may face drastically different risk profiles due to subtle differences in sowing dates, soil orders, canopy density, and irrigation methods. **AASRA transforms reactive guesswork into proactive, evidence-based precision decision support.**

The system strictly decouples **Problem Identification (Diagnostics)** from **Product Recommendation (Prescription)**, ensuring every recommendation provides transparent empirical justification ("Why?"), closes the loop via post-spray visual checks, and demonstrates verifiable economic cash Return on Biostimulant Investment (ROBI).

Pillar / Layer	Primary Core Features	Underlying Agronomic & Mathematical Logic
Layer 1: Field Digital Twin	Feature 1: Satellite Boundary & Soil Auto-Detection	Calculates true acreage via Spherical Excess; detects ICAR Soil Order (Vertisol, Inceptisol, etc.) via GPS to solve farmer uncertainty.
Layer 2: 16-Day Forecast	Feature 2: Compound Climate Stress Feature 3: Month-Dependent Pests	Open-Meteo & MeteoBlue telemetry; calculates VPD, HSI, DSI; aligns calendar month with pest oviposition and degree-day triggers.
Layer 3: Diagnostics	Feature 4: Clinical Diagnostic Mandate (Biotic & Abiotic)	Calculates arrival time (days), biological mechanisms, and potential yield loss in quintals/acre before recommending any product.
Layer 4: Decision Support	Feature 5: Product Compatibility Engine with 3 'Why?' Reasons	Filters 50 CIBRC Syngenta products across 8 criteria; provides field-specific dosage and 48-hour safe meteorological spray window.
Layer 5: Closed-Loop & ROI	Feature 6: Action Tracking Feature 7 & 8: Yield ROI Feature 9: Mandi Optimization	Monitors risk drop (82% -> 51% -> 28%) with leaf appearance checklist; calculates net cash gain and optimizes APMC transport logistics.

Key Operational Insight: AASRA operates on the principle that digital precision agriculture must empower farmers with actionable decision-support rather than serving as an opaque product sales funnel.

2. Feature 1: Personalised Digital Field Twin & ICAR Soil Auto-Detection

The Challenge: Farmers often do not know their scientific soil order, or may incorrectly guess between Black Clay, Alluvial Loam, and Sandy Loam. Furthermore, static coordinates fail to account for field perimeter and true area.

The AASRA Solution:

- **Geodesic Polygon Mapping:** Using high-resolution satellite imagery, the farmer plots field boundary pins. Leaflet calculates true geodesic area using the Spherical Excess formula, accurately computing acres and hectares without distortion.
- **Automated ICAR Soil Classification:** When coordinates are selected, AASRA queries Google Gemini AI grounded in ICAR NBSS&LUP datasets to automatically identify the precise soil order and typical pH range for that specific Indian district:
 - **Maharashtra / MP / Deccan:** *Vertisol (Medium to Deep Black Cotton Clay)*, pH 7.6 - 8.3.
 - **Punjab / Haryana / UP:** *Inceptisol (Deep Alluvial Sandy Loam)*, pH 7.2 - 7.9.
 - **Kerala / Western Ghats:** *Ultisol (Acidic Humid Laterite)*, pH 5.2 - 6.2.
 - **Rajasthan / Gujarat:** *Aridisol (Desert Sandy Loam)*, pH 8.0 - 8.6.
- **Farmer Reassurance & Cross-Check:** The interface displays a live badge '[AI] Soil Auto-Detected: [Soil Type] (pH [X]) • [Texture]' and populates the dropdown with 4–6 local soil variants so the farmer can visually cross-verify and adjust if their specific plot is different.

Key Operational Insight: Boundary coordinates and detected soil chemistry are stored permanently in the database so that subsequent biophysical simulations and pesticide dosage calculations are customized to that exact field boundary.

3. Feature 2: 16-Day Forecast & Compound Climate Stress

AASRA ingests continuous 16-day hyper-local weather telemetry (Open-Meteo & MeteoBlue) for the field's GPS centroid: Daytime Max Temperature (TMax), Nighttime Minimum (TMin), Precipitation (mm), Relative Humidity (RH), Wind Speed, and Solar Radiation.

Atmospheric Vapor Pressure Deficit (VPD) Calculation

$$e_s = 0.6108 * \exp((17.27 * T) / (T + 237.3)) \quad e_a = e_s * (RH_{avg} / 100) \quad VPD = e_s - e_a \text{ [kPa]}$$

Where 'es' is saturated vapor pressure and 'ea' is actual vapor pressure. When VPD > 2.8 kPa, atmospheric moisture demand outstrips root hydraulic capacity.

Compound Climate Stress Index (CS)

$$CS = (0.6 * HSI + 0.4 * DSI) * (1 + 0.3 * HSI * DSI)$$

HSI: Heat Stress Index (TMax & Night Temp vs crop threshold). DSI: Drought Stress Index (VPD, rainfall deficit, dry spell days). The interaction term accounts for catastrophic simultaneous heat + drought.

Nocturnal Heatwave Stress (High Night Temperature - HNT): When night temperatures stay above crop tolerance (e.g., >24.5°C for Soybean, >18°C for Wheat), the plant is forced into accelerated dark mitochondrial respiration, burning sucrose instead of feeding ovaries, causing severe blossom abortion and flower drop.

4. Feature 3: Calendar-Aware Pest & Disease Attack Forecaster

Insect pests and fungal pathogens do not appear at random; their emergence is strictly governed by **calendar month**, **crop phenological stage**, and **micro-climatic incubation windows**.

- **Month & Phenology Coupling:**
 - **Soybean (July - September):** Semilooper & Spodoptera caterpillars strike during breaks in monsoon rainfall when high humidity combines with sunny spells.
 - **Cotton (August - November):** Pink Bollworm strikes at square and boll initiation.
 - **Wheat (December - February):** Yellow/Brown Rust strikes when dense morning dew aligns with 15–22°C temperatures.
- **Dynamic Pan-India AI Grounding:** For any crop in any district, AASRA evaluates real-time temperature, humidity, and consecutive dry days against CIBRC and ICAR entomological databases to output: the exact scientific and local pest name, incubation trigger, time-to-stress in days, early foliar symptoms, and economic threshold level (ETL).

5. Features 4 & 5: Diagnostic Approach & Product Compatibility Engine

AASRA maintains strict operational separation between **Problem Identification (Pillar 1)** and **Product Prescription (Pillar 2)**. The AI never acts as a black-box sales funnel.

Pillar 1: The Diagnostic Mandate

Before considering any product, the engine answers three clinical diagnostic questions:

1. *What stress is developing?* (e.g. Nocturnal Heat Stress, 82% probability)
2. *When will it happen?* (e.g. Expected in 4 Days during peak thermal spike)
3. *What is the severity and yield loss?* (e.g. 18% loss = 2.1 Quintals/Acre)

Pillar 2: The 8-Point Product Compatibility Engine

A candidate product (from a database of 50 CIBRC-registered Syngenta India formulations) is selected only if it satisfies all 8 compatibility criteria:

Criteria	Compatibility Check Rule	Agronomic Rationale
1. Target Stress	Directly targets diagnosed stress	Osmoprotectant for abiotic heat; targeted insecticide for biotic pests.
2. Crop Label	Approved on crop under CIBRC	Prevents illegal off-label use and phytotoxicity risks.
3. Growth Stage	Aligned with vegetative/bloom/pod	Bio-stimulants during bloom; seed treatments strictly pre-sowing.
4. Soil Condition	Matches soil moisture & pH buffer	Ensures root absorption in heavy clays or avoids leaching in sands.
5. Spray Feasibility	Wind <15 km/h, Rain Prob <20%	Prevents chemical droplet drift and immediate rain washoff.
6. Tank-Mix Safety	Zero chemical antagonism	Explicitly forbids mixing with alkaline copper/sulfur that destabilize actives.
7. Trial Efficacy	Peer-reviewed ICAR trial backing	Requires verified trial citations (e.g. >92% efficacy in university trials).
8. Economic Value	Positive net cash return (ROBI > 2.0)	Ensures investment cost is heavily outweighed by protected harvest value.

The 3 'Why?' Reasons — Building Farmer Trust

Farmers distrust arbitrary recommendations. Every AASRA prescription provides three concrete, transparent reasons:

- **Reason 1 (Weather Telemetry):** *'Forecast detects peak night temperature of 26.2°C (>24.5°C threshold), triggering flower abortion.'*
- **Reason 2 (Crop Stage Sensitivity):** *'Your Soybean is at Flowering & Pod Formation (55 DAS), its most heat-vulnerable reproductive window.'*
- **Reason 3 (Biochemical Mode of Action):** *'Quantis supplies betaines and amino acids that up-regulate Heat Shock Proteins (HSP70) to prevent ethylene-induced flower drop.'*

Key Operational Insight: Actionable Prescription: 'Apply Syngenta Quantis® at 250 ml/acre (1.25 L for your 5 Acres) in 150 L water/acre using a fine hollow cone nozzle within the next 48 hours between 6:00 AM - 9:30 AM before heat peaks.'

6. Feature 6: Closed-Loop Intervention & Visual Crop Health Confirmation

Traditional advisory ends when a product is mentioned. AASRA closes the loop by connecting **Prediction -> Recommendation -> Actual Intervention -> Verification**.

- **Digital Field Action Logging:** The farmer confirms application (Yes/No), recording the date, exact dose, method, and treated area in the persistent Field Journal.
- **Dynamic Risk Trajectory:** The biophysical simulator tracks how risk declines over time:
 - **Before Intervention:** Stress Probability is **82%** (Critical Loss Threat).
 - **After 48 Hours:** Stress Probability declines to **51%** as actives stabilize physiology.
 - **After 7 Days:** Stress Probability resolves to **28%** (Normal Baseline).
- **Visual Confirmation Checklist:** The system tells the farmer exactly what the crop *should* look like:
'Ideally, 48–72 hours after applying Quantis, flower retention should rise to >92%. Flowers should not have blackened pedicels, and leaves should remain turgid and erect even under 1:00 PM peak sunlight.'
If the farmer verifies healthy leaves -> Success confirmed. If abnormal symptoms persist -> Automated secondary advisory triggered.

7. Features 7 & 8: Verifiable Economic ROI & Yield Protection Prediction

To eliminate ambiguity, AASRA computes a complete **transparent cash balance sheet** for every recommendation:

Cost / Benefit Component	Calculation Basis (Example: 5 Acre Soybean Farm)	Financial Value (INR)
1. Product Investment	Dealer Price Rs. 1,150/L * 0.25 L/acre * 5 Acres	Rs. 1,438
2. Spraying Labor Cost	Custom hiring rate of Rs. 250 / acre * 5 Acres	Rs. 1,250
3. Water & Mixing Cost	Rs. 50 / acre * 5 Acres	Rs. 250
Total Out-of-Pocket Investment	Product + Labor + Water costs	Rs. 2,938
4. Protected Harvest Yield	2.1 Quintals/Acre prevented loss * 5 Acres	10.5 Quintals
5. Live APMC Mandi Rate	Modal trading rate at Bhopal APMC Mandi	Rs. 4,850 / Quintal
Gross Revenue Protected	10.5 Quintals * Rs. 4,850 / Quintal	Rs. 50,925
Net Cash Profit	Gross Revenue Protected - Total Investment	+ Rs. 47,987
ROBI Multiplier	Gross Revenue Protected / Total Investment	17.3 x ROI

8. Feature 9: Mandi Price Intelligence & Net Realization Optimization

The Transport Pitfall: A neighboring mandi might quote Rs. 100/quintal higher, but transportation, loading fees, and mandi cess could wipe out the additional revenue.

AASRA Net Mandi Optimizer: Calculates the **Net Realized Revenue** for the farmer's specific harvest volume:

Net Mandi Realization Formula

$$\text{Net_Price} = \text{Modal_Price} - ((\text{Distance_km} * \text{Fuel_Trolley_Rate}) / \text{Total_Quintals} + \text{Cess} + \text{Labor})$$

Ranks nearby APMC markets by true cash left in the farmer's pocket after fuel, tractor trolley transit, loading labor, and market fees.

- **Real-Time Modal Price Tracking:** Pulls authentic Agmarknet and APMC records, prioritizing the *modal price* (the most frequent clearing transaction price), rather than misleading mathematical averages.
- **Route & Logistics Guidance:** Recommends which mandi to visit, estimated travel time, fuel cost, and the exact net cash advantage over the nearest local yard.

9. Summary Architecture Matrix (The 9 Core Features)

The following matrix summarizes the 9 integrated features that make AASRA a complete industrial solution:

Feature #	Feature Name	Key Inputs	Core Mathematical / Agronomic Engine	Deliverable Output
1	Digital Field Twin	Pins, Sowing Date, GPS	Spherical Excess + ICAR NBSS&LUP;	Acreage, Soil Order, Cultivar Profile
2	16-Day Forecast	TMax, TMin, RH, Rain, Wind	Tetens VPD + Compound Stress (CS)	16-day daily stress probabilities
3	Pest Forecaster	Calendar month, humidity, rain break	Oviposition degree-days & CIBRC labels	Specific pest attack risk & ETL
4	Stress Diagnostics	Compound stress, phenology stage	Mitochondrial respiration & ABA models	Time to peak stress & Qtl/acre loss
5	Product Compatibility	50 Syngenta products, weather	8-point compatibility + mode of action	Actionable dose & 3 'Why?' reasons
6	Closed-Loop Action	Farmer confirmation, spray date	Biophysical recovery decay trajectory	Risk drop (82%->28%) + leaf checklist
7	Economic ROI	Product price, labor, water	Investment vs protected yield value	ROBI multiplier & net cash balance
8	Yield Protection	ICAR baseline yield, acreage	Phenological loss mitigation curves	Total protected quintals harvest
9	Mandi Optimization	APMC rates, distance, fuel rates	Net realization logistics formula	Best mandi to sell + net profit gain

Key Operational Insight: AASRA is live and deployed in production. Every algorithm described in this document is executable and verifiable via live GPS queries across all 28 states of India at: <https://frontend-phi-flame-21.vercel.app>